

Botany Lab

COURSE CODE: 2129

REVISION: 2021

SEMESTER: 2

Diploma in Wood and Paper Technology

Category

Engineering Science

Periods/Week

3 (L:0, T:0, P:3)

Total Periods

45 per Semester

Credits

No Credit

Course Outcomes (COs)

CO No.	Description	Duration	Cognitive Level
CO1	Interpret Plant diversity and morphology of root, stem and leaves	9 Hours	Applying
CO2	Analyse reproductive morphology and morphology of fruits	9 Hours	Applying
CO3	Identify angiosperms Taxonomy	12 Hours	Applying
CO4	Interpret Plant anatomy (Wood Structure focus)	12 Hours	Applying

Calculations & Measurements

In Botany Lab, calculations are primarily used for microscopic magnification and determining the age of wood samples.

1. Microscopic Magnification

Formula: Total Magnification = Power of Objective Lens $\times$ Power of Ocular (Eyepiece) Lens

Example: If the eyepiece is 10x and the objective lens is 40x:

Calculation: $10 \times 40 = 400x$ total magnification.

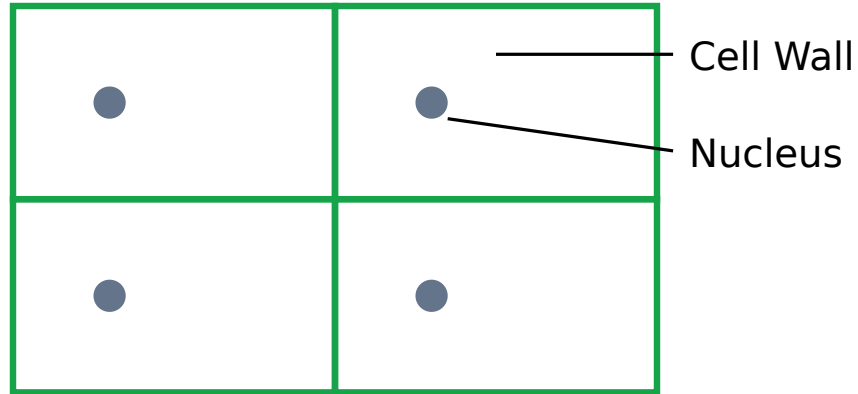
2. Dendrochronology (Tree Age Estimation)

Concept: In temperate and some tropical regions, one growth ring (annual ring) represents one year of growth.

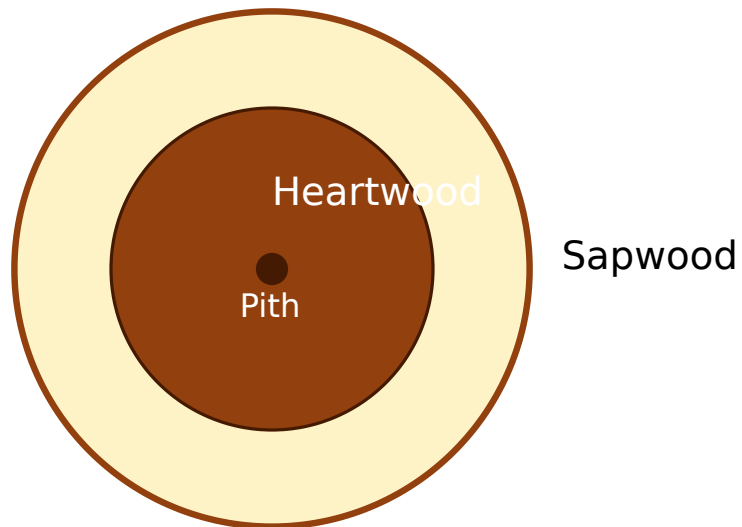
Calculation: Count the number of pairs of light wood (Earlywood) and dark wood (Latewood) from the pith (center) to the vascular cambium (bark side).

Formula: Age (years) $\approx$ Total Number of Annual Rings

Visual Library



Typical rectangular cells of Onion Peel as seen under a light microscope.



Cross-section of a woody stem showing Heartwood (dark, central) and Sapwood (light, outer).

1.1 Microscopic Observation

- **Algae (Spirogyra):** Filamentous green algae with spiral chloroplasts.
- **Fungi (Rhizopus):** Common bread mold; observe sporangia and hyphae.
- **Bryophyta (Moss):** Non-vascular plants; observe gametophyte and sporophyte.
- **Pteridophyta (Nephrolepis):** Ferns; observe sori on the underside of leaflets.

Observe the following organisms: Spirogyra (Algae), Rhizopus (Fungi), Moss (Moss), Nephrolepis (Ferns). Record your observations of the microscopic structures and their functions.

1.2 Morphology of Root, Stem, and Leaves

- **Roots:** Tap root system (Dicot) vs Fibrous root system (Monocot).
- **Stems:** Modifications like Rhizome (Ginger), Tuber (Potato), Bulb (Onion).
- **Leaves:** Venation (Reticulate vs Parallel), Phyllotaxy (Arrangement of leaves).

2.1 Inflorescence & Flowers

Study of how flowers are arranged on the plant axis.

- **Racemose:** Main axis continues to grow (e.g., Mustard).
- **Cymose:** Main axis ends in a flower (e.g., Jasmine).
- **Flower Parts:** Calyx (Sepals), Corolla (Petals), Androecium (Stamens), Gynoecium (Carpels).

2.2 Morphology of Fruits

Type	Description	Example
Simple	From a single ovary	Mango, Tomato
Aggregate	From multiple ovaries of a single flower	Custard Apple
Multiple	From a whole inflorescence	Jackfruit, Pineapple

Study of vegetative and floral characters of key plant families relevant to wood and paper technology.

- **Annonaceae:** Example: Polyalthia (Ashoka tree).
- **Euphorbiaceae:** Example: Ricinus (Castor), Rubber tree.

- **Rutaceae:** Example: Citrus (Lemon).
- **Fabaceae:** Pea family; important for nitrogen fixation.
- **Myrtaceae:** Example: Eucalyptus (Crucial for paper industry).
- **Gramineae (Poaceae):** Grasses, Bamboo (Primary raw material for paper).

ආපෝෂක සම්පත්: පොදු ආපෝෂක සම්පත් වන බැමු (Bamboo - Gramineae), මිරිසවතු (Myrtaceae) ආපෝෂක සම්පත් පිළිබඳව විමර්ශනය කර බලන්න.

4.1 Tissues and Cell Structure

Observation of Onion peel cells to understand basic plant cell structure (Cell wall, Cytoplasm, Nucleus).

4.2 Primary Structure

- **Dicot Stem:** Vascular bundles arranged in a ring; presence of cambium.
- **Monocot Stem:** Scattered vascular bundles; absence of cambium.

4.3 Secondary Thickening (Crucial for Wood Tech)

The increase in girth of the stem due to the activity of vascular cambium and cork cambium.

- **Heartwood (Duramen):** Central, dark-colored, non-functional, durable wood filled with resins/tannins.

- **Sapwood (Alburnum):** Outer, light-colored, functional wood that conducts water.
- **Tyloses:** Outgrowths from parenchyma cells into vessel lumens, often blocking them (common in heartwood).
- **Growth Rings:** Formed by the difference in Earlywood (Spring) and Latewood (Autumn) production.

Module-wise Practice Questions

► **Q1: Differentiate between Heartwood and Sapwood. (CO4)**

► **Q2: What are Tyloses and why are they important in wood technology? (CO4)**

► **Q3: Identify the family of Bamboo and its importance in the paper industry. (CO3)**

► **Q4: What is Dendrochronology? (CO4)**

Practice Model Practical Paper

Time: 3 Hours | Max Marks: 100

1. **Identification (20 Marks):** Identify the given specimens (A-E). Provide one identifying character for each. (Includes Algae, Fungi, Moss, Fern).

2. **Sectioning & Staining (30 Marks):** Take a T.S. of the given plant material (Dicot/Monocot Stem). Stain it and mount it. Draw a neat labeled diagram.
3. **Taxonomy (20 Marks):** Identify the family of the given flowering twig. List the vegetative and floral characters.
4. **Wood Anatomy (20 Marks):** Identify the wood sample provided. Comment on the presence of growth rings, heartwood/sapwood, and hardwood/softwood characteristics.
5. **Viva Voce & Record (10 Marks):** Based on the semester's practical work.

Quick Revision

- **Monocot Stem:** Scattered vascular bundles (like "monkey faces"). No secondary growth.
- **Dicot Stem:** Ring arrangement of vascular bundles. Shows secondary growth.
- **Annual Rings:** One ring = Early wood + Late wood. Used for age determination.
- **Hardwood:** Wood from Angiosperms (contains vessels).
- **Softwood:** Wood from Gymnosperms (lacks vessels, contains tracheids).
- **Phyllotaxy:** Arrangement of leaves on a stem (Alternate, Opposite, Whorled).

Glossary

Cambium

A layer of meristematic tissue responsible for secondary growth (girth increase).

Inflorescence

The arrangement of flowers on a floral axis.

Pith

The central tissue of a stem, usually composed of parenchyma.

Venation

The arrangement of veins in a leaf blade.

Xylem

The vascular tissue responsible for water transport and providing mechanical support (Wood is secondary xylem).

Official Source Citation

This lesson is based on the official syllabus for **Course 2129: Botany Lab** under the **Revision 2021** curriculum provided by the State Institute of Technical Teachers' Training & Research (SITTTR), Kerala.

Source URL: [SITTTR Course 2129](#)

മലയാളം അധികാരസഹായം (Malayalam Support)

ഈ പാഠ്യ പരീക്ഷണ പാഠ്യപുസ്തകം സസ്യങ്ങളുടെ രൂപരേഖ (Morphology),
വർഗ്ഗീകരണം (Taxonomy), സസ്യങ്ങളുടെ ഘടന (Anatomy)
പരീക്ഷണ പാഠ്യപുസ്തകം. സസ്യങ്ങളുടെ ഘടന പരീക്ഷണ പാഠ്യപുസ്തകം
പരീക്ഷണ പാഠ്യപുസ്തകം (Wood) പരീക്ഷണ പാഠ്യപുസ്തകം പരീക്ഷണ പാഠ്യപുസ്തകം
(Heartwood, Sapwood) പരീക്ഷണ പാഠ്യപുസ്തകം പരീക്ഷണ പാഠ്യപുസ്തകം പരീക്ഷണ
പാഠ്യപുസ്തകം. പരീക്ഷണ പാഠ്യപുസ്തകം (Bamboo), പരീക്ഷണ പാഠ്യപുസ്തകം പരീക്ഷണ
പാഠ്യപുസ്തകം പരീക്ഷണ പാഠ്യപുസ്തകം പരീക്ഷണ പാഠ്യപുസ്തകം.